Paper B: engineering validation v2

18 October 2026. Frozen policies; no optimization reruns. New declared participant-level common P/Q mapping, with a separate unchanged historical two-map sensitivity.

Executive scientific verdict

CONDITIONAL GO for the controlled DC accounting claims, with explicit engineering limitations. Historical baseline reconstruction passes. The new common-mapping registry contains 648 cases, and the matched historical-mapping registry contains 648 cases. Across the two campaigns, 1296/1296 cases converge. Under the common map, 294/648 pass the thermal check and 294/648 pass all declared thermal, voltage and simple-Q-bound checks. Peak current loading reaches 278.395%. These are selected stress cases, not annual security certification. Generator-specific balancing allocation and reserve/ramp feasibility remain unidentified.

The debit geometry and Gate 1 annual certificates are unchanged. This new AC experiment changes only the participant reactive embedding; it does not alter A, any action trajectory, background P/Q, generator reference dispatch, ratings, contingency, PF assumption or solve setting. Its common-map choice is a declared modeling assumption, not a reconstruction of undocumented device locations.

Authority, frozen inputs and policies

The publication authority is the hash-verified Gate 3 main manuscript and technical report in `TPWRS_PaperB_Final_Signoff_20261017`, with the new independently reconstructed M1-LP lower bounds and conservative marginal-relief premium. Six authority identities are recorded in `AUTHORITY_IDENTITIES.csv`. The original `Final/specification/MANUSCRIPT.tex` is not substituted. The blocked preflight is retained in its original directory without edits.

The frozen physical case is `2ed58c3ba8dd673cc53996f713acb983126a9df45922afcb2d5adb3c2e34aeb6`, with 45 buses, 114 directional positions, 94 retained electrical states and 8,837 monitored rows per hour. The 2029-2031 evaluations contain 8,760 UTC hours each. All requests, eligibility/action bounds, background injections, A, signed sensitivities, offsets and rating rows retain their original order and hashes.

Annual M1-LP is a relaxation schedule, not an implementable touched-time controller. Annual M2 is the archived selected schedule. Rolling M1/M2, U and C use their original raw implemented actions. EC is the archived retrospective positionwise energy-controlled witness. The raw premium residuals differ slightly from the guarded witnesses printed elsewhere; no raw/guarded trajectories are silently mixed.

Physical metric definitions and objective interpretation

The original objective is S=sum_t Delta_t sum_state,row [abs(f)-F]_+, in MW h row-state. It sums repeated appearances of overloads in multiple contingency states by definition. It is neither unserved energy, load shedding, probability-weighted risk, simultaneous overloaded MW, nor a dispatch-feasibility test.

The complementary hourly diagnostic is V_t=max_state,row [abs(f)-F]_+. Vmax is its annual maximum. Material overload hours count V_t>epsilon at epsilon=1e-4 MW and 1 MW; 1 MW is a chosen reporting convention, not an official security tolerance. Mean, linear 95th/99th percentiles, intact and contingency maxima/hours, distinct physical elements, and worst row metadata are retained. Distinct elements are deduplicated across states. The maximum across states is not a simultaneous real-world loading.

Full-year physical comparisons

year	policy	S (million)	Vmax MW	Hours >1 MW	Mean V MW	P99 V MW
2029	ANNUAL_M1_LP	57.910	2,135.807	8067	634.006	1,945.695
2029	ANNUAL_M2	48.058	1,939.658	8036	585.760	1,793.917
2029	ROLLING_M1	128.614	2,231.711	7362	657.589	2,227.890
2029	ROLLING_M2	100.252	2,229.614	7060	583.191	2,213.542
2029	U	120.435	2,231.711	7235	633.779	2,228.102
2029	C	103.924	2,231.711	7101	591.203	2,226.717
2029	EC	110.551	2,231.711	8057	631.140	2,226.996
2030	ANNUAL_M1_LP	37.782	2,095.206	6536	539.795	1,903.743
2030	ANNUAL_M2	32.238	1,997.237	6531	508.416	1,749.869
2030	ROLLING_M1	100.449	2,229.614	6696	597.055	2,152.448
2030	ROLLING_M2	75.430	2,222.477	6257	519.078	2,080.689
2030	U	92.851	2,229.614	6503	560.710	2,165.867
2030	C	78.052	2,227.260	6280	527.101	2,086.620
2030	EC	84.367	2,227.260	8117	567.068	2,110.258
2031	ANNUAL_M1_LP	23.659	1,887.092	6481	512.021	1,745.830
2031	ANNUAL_M2	18.741	1,738.694	6461	482.077	1,646.389
2031	ROLLING_M1	90.424	2,220.845	6680	590.906	2,123.883
2031	ROLLING_M2	66.157	2,135.807	6219	497.608	2,058.864
2031	U	81.395	2,227.624	6470	547.128	2,098.424
2031	C	68.441	2,135.807	6245	507.954	2,060.530
2031	EC	73.450	2,150.199	8153	555.661	2,074.780

All 21 complete signed-row objective reconstructions agree exactly with the stored independent replay totals. Each object includes all 8,837 rows at all 8,760 hours. Integrated residual and tail/duration metrics measure different outcomes: an annual minimizer of S does not minimize overload-hour count or peak row loading. In particular EC can reduce U's cumulative score while leaving many more hours with a small material violation. A retrospective positionwise scaling may spread residual congestion over time without creating an optimized peak or duration objective. That last sentence describes a possible mechanism; the computed counts and S values are verified, whereas a causal attribution to any one hour is not established.

Figures A-C display the residual/peak relationship, material hours, and unsmoothed exceedance-duration distributions. All axes include their relevant zero baseline. No confidence intervals or independent-year statistical inference are used.

AC model reconciliation and baseline verification

Let sigma_n=+1 for import and -1 for export. The frozen final active map satisfies A[:,n]=sigma_n(C_P[:,n]-pi), and hence p(x)=p_background-C_P diag(sigma)(v-x)+pi sum_n sigma_n(v_n-x_n). Both maps have nonnegative normalized columns. The common version sets C_Q=C_P; the historical version retains its structural C_Q. Their active injections are identical. For lagging PF 0.98, participant Q=C_Q[-sigma*(v-x)*tan(acos(0.98))]; leading reverses sign and unity gives zero participant Q. Column normalization proves total Q is unchanged. An independent participant-by-participant summation reproduces the spatial remapping difference.

The predeclared baseline consists of all three selected states and three PF assumptions at the earliest selected hour in each evaluation year: 27 historical rolling-M2 AC records. Convergence, violation counts, full bus voltages and all monitored loading values match the archive within predeclared tolerances. Archived Q excess, slack P and normalized overload summaries also match. Nine separately executed DC cases reproduce signed affine row flows. The historical archive lacks complete device-level Q vectors, so only its

saved Q summary can be independently matched. New runs retain the complete realized gen/ext-grid Q values and declared limits.

PV generators retain source P, voltage targets and Q limits. Original load/sgen/storage elements are disabled and supplied non-PV injections are represented by synthetic PQ elements. One source external grid remains the slack; it supplies nonlinear losses. This is a fixed-injection AC power flow, not AC-OPF or headroom-constrained redispatch.

The frozen DC F values come from central active-power rating estimates. AC loading uses native current limits (including transformer current loading), not those central MW values. Their source conventions remain unchanged and their row mapping is recorded. The inherited CSV column name `ac_apparent_normalized_thermal_exceedance` denotes J_AC=sum[max(loading_percent/100-1,0)]; with this solver configuration loading is current-based. It is not numerically interchangeable with annual S.

Matched AC outcomes

Map	Policy	N	Converged	Thermal_pass	Voltage_pass	Q_pass	Peak_loading	J_AC_sum
Common v2	C	162	162	78	162	162	217.253	50.573
Common v2	EC	162	162	66	162	162	227.377	58.480
Common v2	ROLLING_M2	162	162	84	162	162	212.695	47.442
Common v2	U	162	162	66	162	162	278.395	54.961
Historical	C	162	162	78	162	162	217.171	50.556
Historical	EC	162	162	66	162	162	227.286	58.453
Historical	ROLLING_M2	162	162	84	162	162	212.634	47.430
Historical	U	162	162	66	162	162	278.279	54.933

The common-map voltage range is 0.965223-1.005115 pu, and its largest recorded simple-Q-bound excess is 0 Mvar. The source bounds are not full capability curves. Nonconvergent cases, if present, remain separate from zero-violation outcomes. A converged power flow may still have severe thermal overload, and a pass of these selected checks does not establish generator headroom or annual AC security.

Policy rankings under each mapping

Map	A	B	A lower J	B lower J	J ties	Both feasible	Only A	Only B	Neither
Common v2	C	ROLLING_M2	0	15	147	78	0	6	78
Common v2	C	U	18	9	135	66	12	0	84
Common v2	EC	U	18	90	54	54	12	12	84
Historical	C	ROLLING_M2	0	15	147	78	0	6	78
Historical	C	U	18	9	135	66	12	0	84
Historical	EC	U	18	90	54	54	12	12	84

For C versus U, the common-map AC score favors C in 18 cases and U in 9, with 135 ties. For C versus ROLLING_M2, the common-map AC score favors C in 0 cases and ROLLING_M2 in 15, with 147 ties. For EC versus U, the common-map AC score favors EC in 18 cases and U in 90, with 54 ties. The EC annual DC improvement therefore does not establish an improvement in these selected AC checks. These counts concern the fixed stress sample only.

Each row covers 162 matched hour/state/PF comparisons. A and B are compared only within the same AC mapping, timestamp, state and PF setting. Lower normalized J uses a deadband of 1e-4; loading uses 1e-4 percentage point, voltage deviation 1e-6 pu and Q excess 1e-6 Mvar. All-feasibility counts apply the declared voltage 0.95-1.05 pu, thermal 100%+1e-6 tolerance and Q excess <=1e-6 Mvar. Repeated PFs and contingencies are deterministic sensitivities, not independent trials.

Mapping sensitivity, kept separate from policy effects

Policy	N	Common_lower_J	Historical_lower_J	J_ties	Max_abs_J_change	Max_abs_loading_change_pp	Changed_feasibility
C	162	13	21	128	0.002	0.150	0
EC	162	5	24	133	0.002	0.157	0
ROLLING_M2	162	13	15	134	0.002	0.061	0
U	162	13	31	118	0.002	0.151	0

A	B	Pairs	Strict_J_reversals	Changed_J_class	Changed_feasibility_class
C	U	162	0	0	0
C	ROLLING_M2	162	0	0	0
EC	U	162	0	0	0

Policy	Convergence_changes	Thermal_status_changes	Voltage_status_changes	Q_status_changes	Peak_changes_over_1pp
ROLLING_M2	0	0	0	0	0
U	0	0	0	0	0
C	0	0	0	0	0
EC	0	0	0	0	0

Differences above are common minus historical at identical policy actions. Every unity-PF result is unaffected within 0 in the checked metric values. Nonunity changes reflect spatial participant-Q allocation; they do not change total participant Q or active restrictions. Mapping sensitivity is reported for all policies, including the new premiums, without pooling the two mappings in a policy ranking. Changes in the policy win/loss/tie counts between the two policy-comparison blocks show whether ranking conclusions depend on the mapping.

Generator headroom and fixed balancing

year	policy	Conditional hours	Buses	Up excess MW	Down excess MW
2029	ANNUAL_M1_LP	8760	31	1,265.500	1,230.456
2029	ANNUAL_M2	8760	31	1,220.743	1,204.777
2029	ROLLING_M1	8760	31	1,441.256	1,236.436
2029	ROLLING_M2	8760	31	1,435.205	1,201.256
2029	U	8760	31	1,442.736	1,236.925
2029	C	8760	31	1,441.917	1,204.395
2029	EC	8760	31	1,441.917	1,210.770
2030	ANNUAL_M1_LP	8760	31	1,159.096	1,160.213
2030	ANNUAL_M2	8760	31	1,047.894	1,160.213
2030	ROLLING_M1	8760	31	1,384.845	1,163.735
2030	ROLLING_M2	8760	31	1,313.863	1,163.735
2030	U	8760	31	1,384.845	1,163.735
2030	C	8760	31	1,321.623	1,163.735
2030	EC	8760	31	1,321.623	1,166.686
2031	ANNUAL_M1_LP	8760	37	1,628.294	1,167.806
2031	ANNUAL_M2	8760	37	1,477.533	1,167.806
2031	ROLLING_M1	8760	37	1,877.359	1,174.415
2031	ROLLING_M2	8760	37	1,810.365	1,174.415
2031	U	8760	37	1,810.365	1,174.415
2031	C	8760	37	1,810.365	1,174.415
2031	EC	8760	37	1,818.852	1,185.131

The fleet-only test fails in every hour of every policy-year. This count is structurally driven in part by balancing shares at buses 4, 5, 17, 32, 34, where the reconstructed native generator/interchange fleet contains no device; their combined balancing share is 6.910%. The model also contains separate renewable/load components, so absence from that native fleet does not prove absence of all physical resources. The all-hour failure must not be interpreted as an identified generator violation or as a policy ranking. It demonstrates that the fixed balancing key cannot automatically be interpreted as redispatch of only the inventoried native fleet.

These are CONDITIONAL bus-aggregate necessary-condition results, not identified generator-specific headroom violations. The diagnostic first separates background generation and boundary devices from participant demand/generation and renewable curtailment. It then asks whether assigning the complete fixed balancing component to that same native generator/interchange fleet, with the other resource components fixed, would exceed summed native P bounds. The separate headroom report states the condition, recovered data and limitations.

The actual A supplies bus shares but no generator-specific allocation. For two generators at one bus, adjustments (d,0) and (0,d) can produce the same bus injection and different device headroom use. Therefore bus injections alone cannot identify Delta P_g. If a fleet-only aggregate bound fails, no allocation within that conditional fleet can realize the same aggregate; this follows by summing its device bounds. It does not rule out other unmodeled load/renewable allocations. Passing an aggregate bound cannot establish reserves, market availability, ramps or individual deliverability.

Information timing and nonanticipativity

The rolling benchmark uses realized controlled-scenario inputs over a deterministic 48-hour window and commits 24 hours. It is distinct from full-year annual foresight. The marginal score uses the requested-state derivative and no optimized future action, but its future-hour requested/background inputs are not authenticated forecast vintages. The card/thresholds are frozen across held-out years; each year's bucket assignments vary with its scenario inputs.

A prospective schedule and rate would have to be measurable with respect to information at the applicable announcement time. Forecast products, issue-time stamps, revisions, network availability and contract announcement/settlement rules would need authentication. No 07:00 implementation, forecasting experiment or prospective controller is validated. Finite-window perfect information is not a general optimal-value upper bound on all prospective controllers; comparable global horizons, domains and nonanticipativity assumptions would be required.

Numerical QC, reproducibility and scientific scope

The independent QC aggregates every persisted AC voltage, loading and Q-device row and checks all case metrics and paired keys. It recomputes all hourly diagnostic statistics, verifies alternate physical algebra on a declared sample, and checks all 525 critical row flows directly. Complete objective agreement is independently tied to the stored fixed-action replay identities. The preflight, physical, AC, headroom, timing and independent-QC status files distinguish execution success from physical feasibility.

All new work is in the version-2 directory. Per-case AC and per-object physical checkpoints permit restarting without replacing completed results. The default no-argument PyCharm stage is PREFLIGHT. No annual/rolling optimization, card fitting, trajectory repair, generator/rating change, IEEE/Zenodo upload or modification of the old blocked preflight occurs. Consumed files and sealed Gate 1/Gate 3 identities are checked again before final sealing.

Impact on the TPWRS manuscript and remaining limitations

The contribution remains the entitlement-domain analysis, exact relaxed-hybrid projection and instance-specific strict annual numerical separation. Different feasible domains do not imply different optimal residuals in every instance. The new AC exercise is an engineering qualification, not a new debit rule or proof of contract superiority. C improves on U in the recorded rolling DC experiment, while ordinary M2 remains better in every held-out year. EC remains a common-feasible retrospective control, not a deployable controller or proof that a differentiated contract is necessary.

Keep one concise main-paper engineering paragraph with the declared common-map assumption and the most consequential thermal/headroom/timing limitation. Put detailed tables and mapping sensitivity in the technical report. Proposed source-grounded LaTeX is provided separately; the locked ten-page manuscript is not overwritten. No DOI for the new report is invented. Submission remains CONDITIONAL GO for the controlled-model claims, conditional on accurate model/rating/chronology descriptions and retention of adverse engineering findings. Secure operational dispatch, annual AC feasibility and prospective contract superiority remain unproved.

Evidence-to-code map

Finding	Isolated script	Frozen input family	New result
Baseline identity	stage_preflight.py	selected_ac and engineering_selection	BASELINE_AC_RECONSTRUCTION.csv
Full-year metrics	stage_physical.py	physical_case arrays and saved actions	ANNUAL_PHYSICAL_METRICS.csv
Mapping and AC comparisons	stage_ac.py	ac_inputs, native network, original sample	MATCHED_AC_CASES.csv and AC_MAPPING_SENSITIVITY.csv
Balancing scope	stage_headroom.py	device registry, background caches, A	BALANCING_FEASIBILITY_BY_POLICY.csv
Information timing	stage_information.py	rolling source, frozen card and experiment records	INFORMATION_STRUCTURE_AND_0700_AUDIT.md
Independent QC	stage_qc.py	frozen arrays and completed row tables	INDEPENDENT_NUMERICAL_QC.json

The full absolute paths, hashes and result links appear in SOURCE_AND_RESULT_TRACEABILITY.csv and the consumed-source inventory; the reproducibility guide defines each root and safe stage usage.

Traceable engineering figures

Figure A. Residual and maximum retained-row exceedance.

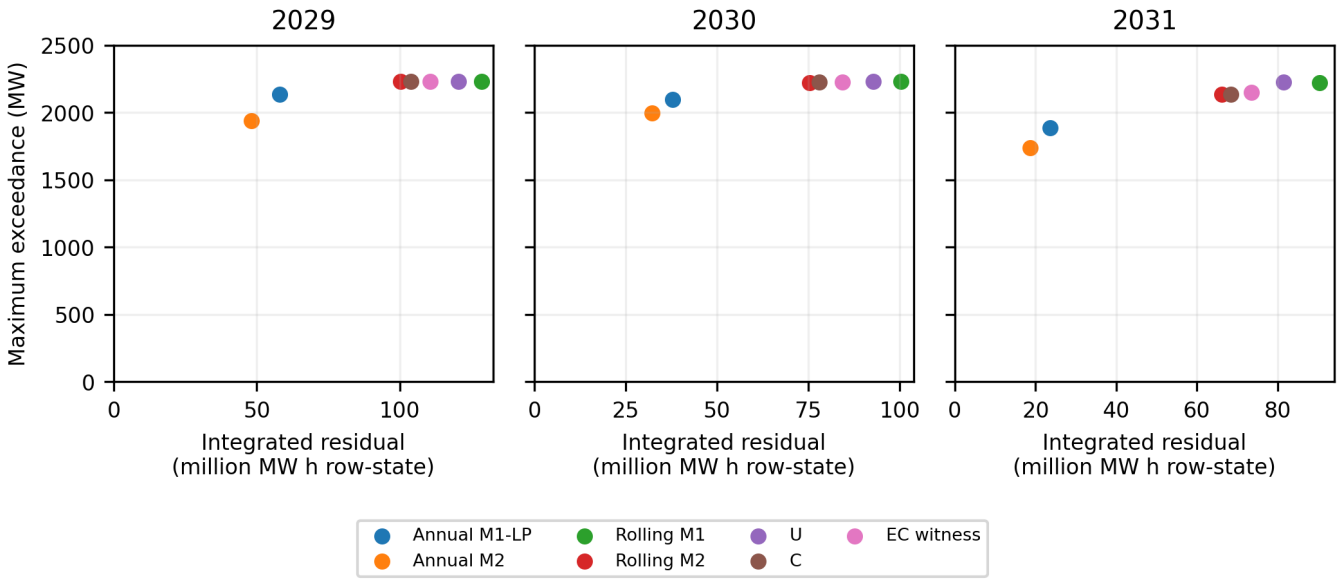


Figure B. Full-year material violation hours.

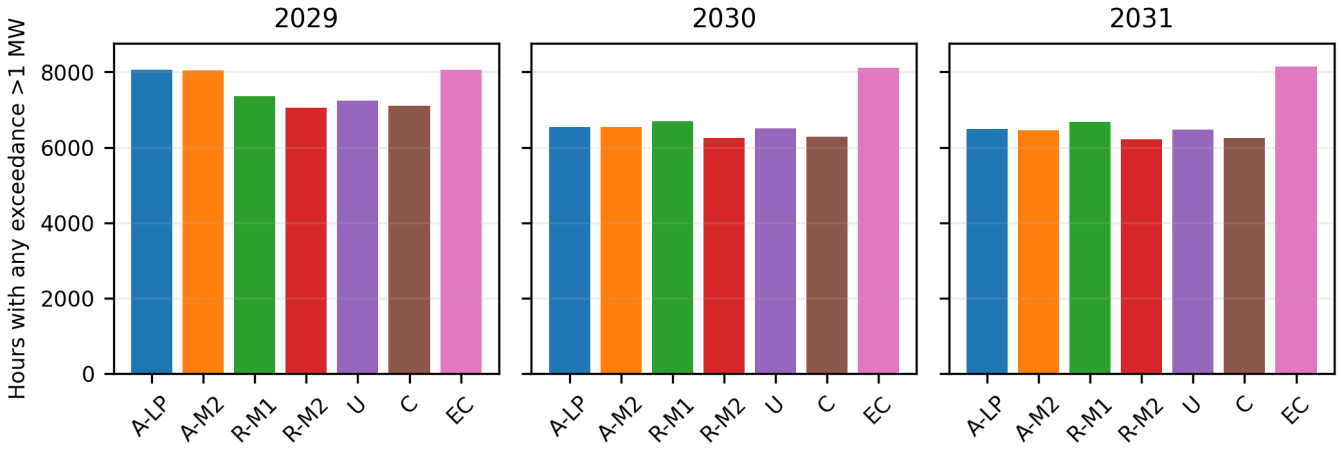


Figure C. Hourly maximum exceedance-duration curves.

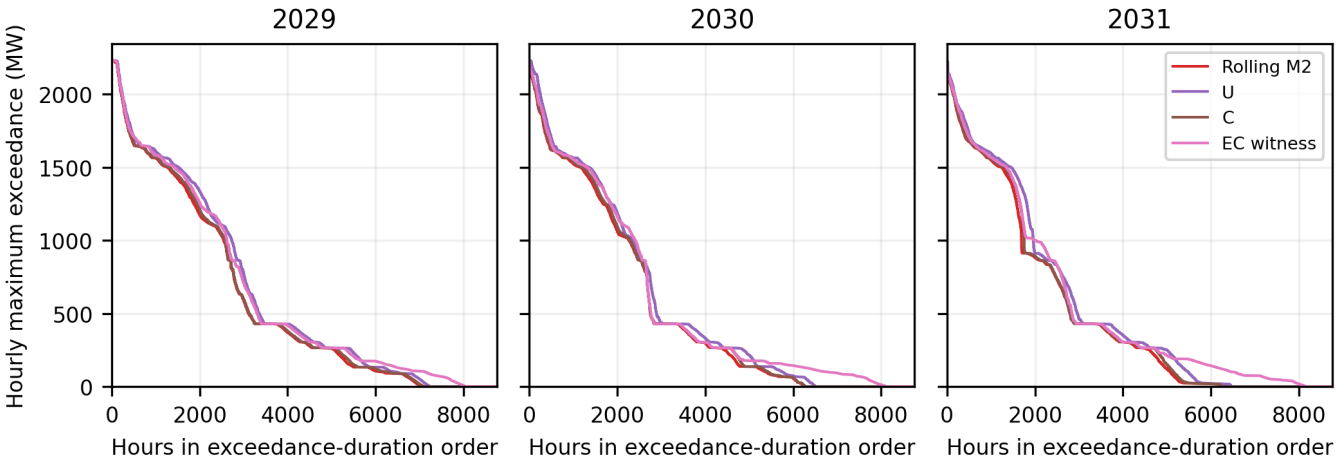


Figure D. Matched selected-case AC overload, by mapping.

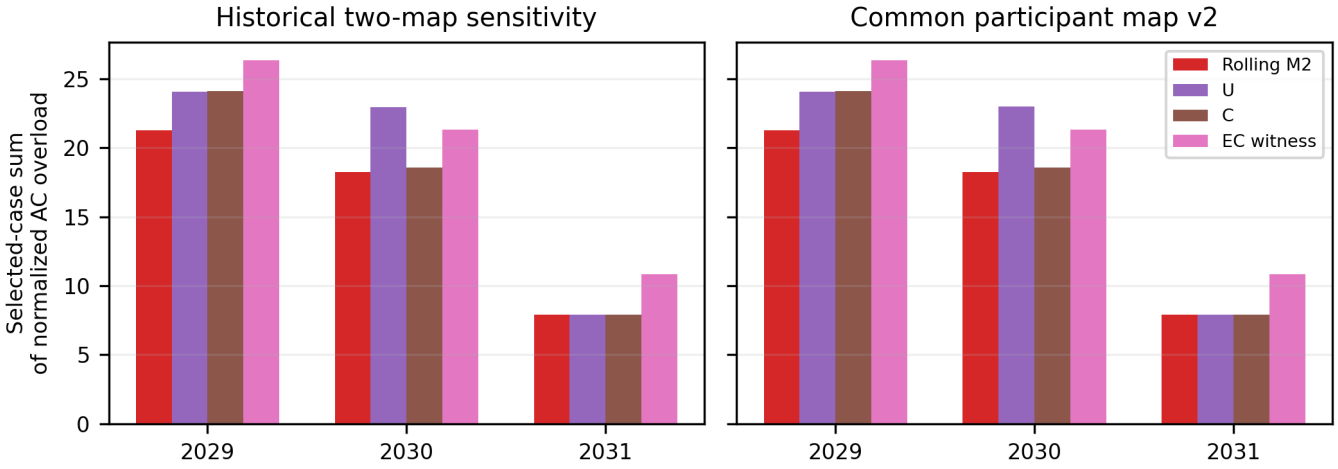


Figure E. Selected-case thermal pass and peak loading.

